

GRATIOT, MI, USA

WMO: 725405

Lat: 43.322N

Lon: 84.688W

Elev: 228

StdP: 98.62

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.2	-22.4	0.5	-18.7	-19.5	0.7	-14.8	10.2	-5.4	9.1	-4.3	1.3	240	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	31.8	22.4	30.1	21.9	28.6	20.8	24.6	28.9	23.5	27.7	22.6	26.5	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.2	26.7	22.3	17.5	25.9	21.3	16.4	24.8	75.6	28.5	71.1	27.4	67.5	26.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.4	-23.7	33.5	3.3	1.6	-26.0	34.7	-27.9	35.6	-29.8	36.5	-32.1	37.6		
			-23.9	26.1	3.2	1.3	-26.2	27.0	-28.0	27.8	-29.8	28.5	-32.1	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-5.3	-4.7	1.0	7.7	14.1	19.0	21.3	20.1	16.3	9.9	3.7	-2.2
	DBStd	10.57	6.06	5.91	6.18	5.02	4.69	3.80	3.33	3.00	4.18	4.76	4.88	5.32
	HDD10.0	1927	473	411	287	102	13	0	0	0	3	63	196	379
	HDD18.3	3871	731	643	538	319	147	36	11	16	88	265	438	637
	CDD10.0	1375	0	0	8	33	142	272	349	313	191	59	8	1
	CDD18.3	278	0	0	1	1	17	58	102	71	26	3	0	0
	CDH23.3	2806	0	0	4	26	191	581	1039	654	278	33	0	0
	CDH26.7	816	0	0	1	3	42	174	344	174	73	4	0	0
Wind	WSAvg	3.1	3.5	3.5	3.5	3.9	3.3	2.7	2.4	2.2	2.4	3.1	3.4	3.5
Precipitation	PrecAvg	790	45	35	54	74	79	81	66	90	86	65	69	51
	PrecMax	1066	99	111	155	186	201	172	179	255	414	152	160	125
	PrecMin	566	5	1	6	22	8	18	9	28	1	5	13	7
	PrecStd	122	25	22	28	31	43	38	35	44	65	34	37	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	13.5	22.5	27.0	30.1	32.5	33.7	32.6	31.8	27.1	20.2	13.9
		MCWB	10.2	10.4	16.2	16.4	20.9	22.6	23.5	22.7	20.4	20.4	13.8	12.1
	2%	DB	7.4	8.0	17.3	22.7	27.7	30.9	32.1	30.4	28.7	23.7	16.3	9.1
		MCWB	5.7	5.9	12.4	14.2	18.8	21.9	23.2	22.3	20.4	17.2	11.9	7.8
	5%	DB	4.1	5.2	13.0	20.0	25.7	28.7	30.2	28.5	26.7	21.0	13.7	7.1
		MCWB	2.7	3.2	9.5	12.9	17.7	20.6	22.4	21.2	19.5	16.2	10.5	5.8
	10%	DB	2.4	2.8	10.2	17.2	22.9	27.0	28.4	27.2	24.2	18.2	11.5	4.1
		MCWB	1.5	1.3	7.0	11.2	16.3	19.8	21.1	20.3	18.5	14.4	8.8	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	10.9	16.9	18.9	22.5	24.7	26.4	25.5	24.1	21.5	15.4	12.6
		MCDB	11.0	12.1	21.7	24.6	27.6	29.7	31.2	30.3	27.9	25.6	18.3	13.9
	2%	WB	6.0	6.6	13.3	16.0	20.8	23.3	25.0	24.0	22.5	19.3	13.0	8.1
		MCDB	7.2	8.6	16.4	19.9	24.9	28.1	29.5	27.8	25.9	21.6	16.0	8.9
	5%	WB	3.2	3.5	10.6	14.0	19.3	22.1	23.8	22.8	21.3	17.1	11.0	5.8
		MCDB	4.2	5.0	13.2	18.7	23.3	26.8	28.1	26.4	24.3	19.5	13.7	7.0
	10%	WB	1.5	1.4	7.0	11.8	17.8	21.0	22.8	21.8	19.8	15.0	8.8	3.2
		MCDB	2.3	2.8	9.3	16.4	21.5	25.0	26.8	25.3	22.9	18.0	11.1	4.3
Mean Daily Temperature Range		MDBR	7.4	8.6	9.8	11.5	11.8	12.0	12.5	12.2	12.7	10.8	8.5	6.7
	5% DB	MCDBR	8.2	9.7	14.1	16.2	14.4	13.5	13.7	13.8	15.1	14.7	12.2	8.9
		MCWBR	7.3	7.7	9.5	8.9	7.3	6.3	6.5	6.8	7.8	8.7	8.6	7.9
		MCDBR	8.0	9.1	12.8	13.6	12.2	11.6	11.5	11.6	11.6	12.2	11.1	8.8
	5% WB	MCWBR	7.3	7.7	9.6	9.2	7.4	6.4	6.4	6.7	7.1	8.8	8.9	8.1
Clear-Sky Solar Irradiance	taub	0.289	0.327	0.330	0.370	0.397	0.422	0.411	0.405	0.377	0.338	0.309	0.286	
	taud	2.360	2.241	2.382	2.300	2.275	2.238	2.274	2.288	2.359	2.469	2.493	2.424	
	Ebn at Noon	843	858	911	895	876	852	857	850	848	845	820	816	
	Edh at Noon	79	106	105	124	131	137	130	124	107	85	70	68	
All-Sky Solar Radiation	RadAvg	1.39	2.27	3.60	4.57	5.37	6.04	6.11	5.20	4.17	2.62	1.59	1.11	
	RadStd	0.12	0.27	0.39	0.40	0.50	0.36	0.31	0.25	0.40	0.28	0.21	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page